REQUEST FOR PROPOSAL
RFP #: HK – F3.H3

TITLE: HURRICANE READINESS PROJECT – PHASE 3
CLOSING DATE AND TIME: February 13. 2026 @ 5:00 PM

Hurricane Readiness Project – Phase 3: HK – F3.H3

Purpose
By responding to this Request for Proposal (RFP), the Proposer agrees that s/he has read and understood all documents within this RFP package.

Submission Details
Responders to this RFP should supply:
- A business report **up to 5 pages** (not including cover page, table of contents, or any needed appendix), including any supporting plots and tables.
- The commented code used to produce the results in separate file(s).

The report should address **all points described in the "Objective" section** below.

The report should be returned in the following way:
- Electronic – Moodle submission on AA502 website

Background
Several hurricanes struck the gulf area and resulted in severe casualty and property damage. One of the major defenses is to maintain and coordinate the pump operations during a critical 48 hour period (or over 4 high tides). The Steering Committee of the Center for Risk Management (hereafter the "Committee") is conducting an analysis for the pump stations in the gulf coast area to help better prepare for future hurricanes.

Pumps may or may not fail during this critical 48-hour window in a hurricane. If they fail, flood waters begin to rise too quickly which leads to catastrophic damage to homes and businesses in the area and severe loss of life. Pumps can failfor a variety of reasons during a hurricane. The four currently tracked by the Committee are the following:
1. Flood – overflow or accumulation of an expanse of water that submerges the pump station.
2. Motor – mechanical failure of the pump motor.
3. Surge – onshore gush of water usually associated with levee or structural failure.
4. Jammed – accumulation of trash or landslide materials which leads to water not getting to the pump.

For further information on how New Orleans specifically has adjusted since hurricane Katrina with the upgrades of such pumps please see the following article from the Wall Street Journal:
https://www.wsj.com/articles/how-new-orleans-fortified-itself-against-water-11562981176

Objective
The scope of services in this third phase includes the following:
- For this phase use **the entire** data set.
- Build a Cox proportional hazard model for **motor failure** with main effects only using backward elimination.
    - The Committee currently uses a significance level of **0.03** and backward selection.
    - Feel free to use a different significance level and different selection technique aslong as

  - you defend your reason.
  - List a table of significant variables ranked by their p-value.
  - Interpret the effects of the most significant variable.
- The Army Corp of Engineers and the Committee are interested in finding out if this type of failure ismore likely to occur if the motor is constantly running for 12 hours blocks.
  - Use the information provided in the variables H1 through H48 to help create a feature for capturing when a pump has been constantly running for 12 hours (this new data set should have one row per hour for the length of a time a pump is running). Add this new variable to the previous model and report if it has a significant impact on the hazard of motor failure
  - Make sure to test any needed assumptions you may have for this model and report these findings.
  - (HINT: You will want to use the H1-H48 variables to build a variable to analyze the constant running of the motor. You will need to make some assumptions (be sure to state any assumptions you make). Keep in mind that pumps can be turned on and off during this timeframe).

Data Provided

The following data set is provided for the proposal:

- The data set **hurricane** contains 770 observations and 59 variables.
  - All of the pumps have a reason for failure in the variable **reason**:
    - 0 – no failure
    - 1 – flood failure
    - 2 – motor failure
    - 3 – surge failure
    - 4 – jammed failure
  - There are 56 variables describing the pump's factors that potentially influence the survivability of the pump stations (not all pumps have each characteristic, but some characteristics are available through upgrade or maintenance).

Name	*Model Role*	*Description*
AGE	Input	Difference between the installation and the current date
BACKUP	Input	Redundant system used to protect the station from flooding when the main pump is not operating (UPGRADE AVAILABLE – $100K)
BRIDGECRANE	Input	Allow vertical access to equipment and protecting materials (UPGRADE AVAILABLE – $50K)
ELEVATION	Input	Elevation of the pump station that can be altered by 1 foot by maintenance (MAINTANENCE AVAILABLE – $10K/FT)
GEAR	Input	Gear box used to make the pumps stronger and faster (UPGRADE AVAILABLE – $75K)
H1 – H48	Input	Pumping status during a 48 hour emergency reported by pump stations – accuracy of pump status not guaranteed to be error free
HOUR	Target	Hour that the pump failed or was censored
REASON	Strata	Reason for pump failure (recorded as 0, 1, 2, 3, or 4)
SERVO	Input	Servomechanism is used to provide control of a desired operation through the Supervisory control and data acquisition (SCADA) system (UPGRADE AVAILABLE – $150K)
SLOPE	Input	Surrounding ravine slope of the pump station
SURVIVE	Target	If the pump survived the hurricane without failure
TRASHRACK	Input	Used for protecting hydraulic structures against the inlet of debris, of vegetation, urban or industrial trash (UPGRADE AVAILABLE – $80K)